

Lab07_EduardoHsu

Eduardo Hsu 277142

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(DBI)
```

Warning: pacote 'DBI' foi compilado no R versão 4.6.1

```
library(RSQLite)
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.6.1

```
conn <- dbConnect(SQLite(), "uwmadison.sqlite3")
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation="STAT");

```
dbGetQuery(conn, 'SELECT COUNT(DISTINCT instructors.name) as Total
FROM subjects
JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code
JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid
JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid
JOIN teachings
  ON sections.uuid = teachings.section_uuid
JOIN instructors
  ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = "STAT"')
```

Total
1 377

```
dbGetQuery(conn, 'SELECT DISTINCT instructors.name
FROM subjects
JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code
JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid
JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid
JOIN teachings
  ON sections.uuid = teachings.section_uuid
JOIN instructors
  ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = "STAT"
LIMIT 5') #LIMIT para não poluir o PDF
```

	name
1	MINJING TAO
2	DONALD PORTER
3	SHENG WANG
4	KUNLING HUANG
5	DONG XIA

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

```
dbGetQuery(conn, '
SELECT
  courses.name AS disciplina,
  (
    4.0 * grade_distributions.a_count +
    3.5 * grade_distributions.ab_count +
    3.0 * grade_distributions.b_count +
    2.5 * grade_distributions.bc_count +
    2.0 * grade_distributions.c_count +
    1.0 * grade_distributions.d_count +
    0.0 * grade_distributions.f_count
```

```

    )
    /
    (
        grade_distributions.a_count +
        grade_distributions.ab_count +
        grade_distributions.b_count +
        grade_distributions.bc_count +
        grade_distributions.c_count +
        grade_distributions.d_count +
        grade_distributions.f_count
    ) AS gpa

FROM subjects

JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code

JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid

JOIN courses
    ON course_offerings.course_uuid = courses.uuid

JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid

JOIN grade_distributions
    ON course_offerings.uuid = grade_distributions.course_offering_uuid
    AND sections.number = grade_distributions.section_number

WHERE subjects.abbreviation = "STAT"
AND (
    grade_distributions.a_count +
    grade_distributions.ab_count +
    grade_distributions.b_count +
    grade_distributions.bc_count +
    grade_distributions.c_count +
    grade_distributions.d_count +
    grade_distributions.f_count
) > 0
LIMIT 5') #LIMIT para não poluir o PDF

```

	disciplina	gpa
1	Introductory Statistics for Engineers	2.959770
2	Introductory Statistics for Engineers	3.060976
3	Introductory Statistics for Engineers	2.916667
4	Introductory Statistics for Engineers	3.000000
5	Introductory Statistics for Engineers	2.839286

```

dbGetQuery(conn, '
SELECT
    instructors.name AS professor,

    SUM(
        (
            4.0 * grade_distributions.a_count +
            3.5 * grade_distributions.ab_count +
            3.0 * grade_distributions.b_count +
            2.5 * grade_distributions.bc_count +
            2.0 * grade_distributions.c_count +
            1.0 * grade_distributions.d_count +
            0.0 * grade_distributions.f_count
        ))
    /
    SUM((
        grade_distributions.a_count +
        grade_distributions.ab_count +
        grade_distributions.b_count +
        grade_distributions.bc_count +
        grade_distributions.c_count +
        grade_distributions.d_count +
        grade_distributions.f_count
    ))
    ) AS gpa_medio

FROM subjects

JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code

JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid

JOIN courses

```

```

        ON course_offerings.course_uuid = courses.uuid

JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid

JOIN teachings
    ON sections.uuid = teachings.section_uuid

JOIN instructors
    ON teachings.instructor_id = instructors.id

JOIN grade_distributions
    ON course_offerings.uuid = grade_distributions.course_offering_uuid
    AND sections.number = grade_distributions.section_number

WHERE subjects.abbreviation = "STAT"

GROUP BY instructors.name
HAVING SUM(
    grade_distributions.a_count +
    grade_distributions.ab_count +
    grade_distributions.b_count +
    grade_distributions.bc_count +
    grade_distributions.c_count +
    grade_distributions.d_count +
    grade_distributions.f_count
) > 0

ORDER BY gpa_medio DESC
LIMIT 5') #LIMIT para não poluir o PDF

```

	professor	gpa_medio
1	YAJUAN SI	4.000000
2	SRIKANTHMADHAVAN ARAVAMUTHAN	4.000000
3	GUANHUA CHEN	4.000000
4	GRACE WAHBA	3.971154
5	ZIFENG ZHAO	3.954545

```

dbGetQuery(conn, '
SELECT
    instructors.name AS professor,

```

```

SUM(
    (
        4.0 * grade_distributions.a_count +
        3.5 * grade_distributions.ab_count +
        3.0 * grade_distributions.b_count +
        2.5 * grade_distributions.bc_count +
        2.0 * grade_distributions.c_count +
        1.0 * grade_distributions.d_count +
        0.0 * grade_distributions.f_count
    ))
/
SUM((
    grade_distributions.a_count +
    grade_distributions.ab_count +
    grade_distributions.b_count +
    grade_distributions.bc_count +
    grade_distributions.c_count +
    grade_distributions.d_count +
    grade_distributions.f_count
))
) AS gpa_medio

FROM subjects

JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code

JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid

JOIN courses
    ON course_offerings.course_uuid = courses.uuid

JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid

JOIN teachings
    ON sections.uuid = teachings.section_uuid

JOIN instructors
    ON teachings.instructor_id = instructors.id

```

```

JOIN grade_distributions
  ON course_offerings.uuid = grade_distributions.course_offering_uuid
  AND sections.number = grade_distributions.section_number

WHERE subjects.abbreviation = "STAT"

GROUP BY instructors.name
HAVING SUM(
  grade_distributions.a_count +
  grade_distributions.ab_count +
  grade_distributions.b_count +
  grade_distributions.bc_count +
  grade_distributions.c_count +
  grade_distributions.d_count +
  grade_distributions.f_count
) > 0

ORDER BY gpa_medio
LIMIT 5') #LIMIT para não poluir o PDF

```

	professor	gpa_medio
1	JAMES D KUELBS	2.637500
2	ALEXANDER FISH	2.638889
3	GARY HOWARD SCHROEDER	2.729032
4	SAMUEL STECHMANN	2.741379
5	FANG FANG	2.771127

```

dbGetQuery(conn, '
SELECT
  courses.name AS disciplina,
  courses.number AS numero,

  SUM(
    (
      4.0 * grade_distributions.a_count +
      3.5 * grade_distributions.ab_count +
      3.0 * grade_distributions.b_count +
      2.5 * grade_distributions.bc_count +
      2.0 * grade_distributions.c_count +
      1.0 * grade_distributions.d_count +
      0.0 * grade_distributions.f_count
    )
  )

```

```

    ))
    /
    SUM((
        grade_distributions.a_count +
        grade_distributions.ab_count +
        grade_distributions.b_count +
        grade_distributions.bc_count +
        grade_distributions.c_count +
        grade_distributions.d_count +
        grade_distributions.f_count
    )
) AS gpa_medio

FROM subjects

JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code

JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid

JOIN courses
    ON course_offerings.course_uuid = courses.uuid

JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid

JOIN grade_distributions
    ON course_offerings.uuid = grade_distributions.course_offering_uuid
    AND sections.number = grade_distributions.section_number

WHERE subjects.abbreviation = "STAT"

GROUP BY courses.name, courses.number
HAVING SUM(
    grade_distributions.a_count +
    grade_distributions.ab_count +
    grade_distributions.b_count +
    grade_distributions.bc_count +
    grade_distributions.c_count +
    grade_distributions.d_count +
    grade_distributions.f_count

```

```
) > 0
ORDER BY gpa_medio DESC
LIMIT 5') #LIMIT para não poluir o PDF
```

	disciplina	numero	gpa_medio
1	Data Science Practicum	628	4.000000
2	Empir Proc&Semiparmtrc Infernc	834	4.000000
3	Nonparametric Statistics and Machine Learning Methods	841	4.000000
4	Sample Survey Theory and Method	811	4.000000
5	Experimental Design I	803	3.984127

```
dbGetQuery(conn, '
SELECT
  courses.name AS disciplina,
  courses.number AS numero,

  SUM(
    (
      4.0 * grade_distributions.a_count +
      3.5 * grade_distributions.ab_count +
      3.0 * grade_distributions.b_count +
      2.5 * grade_distributions.bc_count +
      2.0 * grade_distributions.c_count +
      1.0 * grade_distributions.d_count +
      0.0 * grade_distributions.f_count
    )
  ) /
  SUM((
    grade_distributions.a_count +
    grade_distributions.ab_count +
    grade_distributions.b_count +
    grade_distributions.bc_count +
    grade_distributions.c_count +
    grade_distributions.d_count +
    grade_distributions.f_count
  ))
  ) AS gpa_medio

FROM subjects

JOIN subject_memberships
```

```

    ON subjects.code = subject_memberships.subject_code

JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid

JOIN courses
    ON course_offerings.course_uuid = courses.uuid

JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid

JOIN grade_distributions
    ON course_offerings.uuid = grade_distributions.course_offering_uuid
    AND sections.number = grade_distributions.section_number

WHERE subjects.abbreviation = "STAT"

GROUP BY courses.name, courses.number
HAVING SUM(
    grade_distributions.a_count +
    grade_distributions.ab_count +
    grade_distributions.b_count +
    grade_distributions.bc_count +
    grade_distributions.c_count +
    grade_distributions.d_count +
    grade_distributions.f_count
) > 0
ORDER BY gpa_medio
LIMIT 5') #LIMIT para não poluir o PDF

```

	disciplina	numero	gpa_medio
1	Introduction to the Theory of Probability	431	2.886374
2	Linear Programming Methods	525	2.949212
3	Introduction to Probability and Mathematical Statistics I	309	3.026515
4	Introduction to Stochastic Processes	632	3.044066
5	Introduction to Probability and Mathematical Statistics II	310	3.057717

Quem é o professor mais difícil?

O professor considerado mais difícil, ou seja, com menor gpa médio é JAMES D KUELBS.

Quem é o professor mais fácil?

O professor considerado mais fácil, ou seja, com maior gpa médio é um empate entre YAJUAN SI, SRIKANTHMADHAVAN ARAVAMUTHAN e GUANHUA CHEN.

Qual é a disciplina mais difícil?

A disciplina considerada mais difícil, isto é, com menor gpa médio é Introduction to the Theory of Probability

Qual é a disciplina mais fácil?

A disciplina considerada mais fácil, isto é com maior gpa médio é um empate entre Data Science Practicum, Empir Proc&Semiparmtrc Infernc, Nonparametric Statistics and Machine Learning Methods e Sample Survey Theory and Method.

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```